Department of Electrical and Computer Engineering
North South University

DIRECTED RESEARCH 498R

SignSpeak Bangla: Word-Based Dynamic Gesture Recognition System

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LETTER OF TRANSMITTAL

December, 2024

To

Dr. Mohammad Abdul Matin

Chairman,
Department of Electrical and Computer Engineering
North South University, Dhaka

Subject: **Submission of Directed Research Report on "SignSpeak Bangla: Word-Based Dynamic Gesture Recognition System"**

Dear Sir,

With due respect, we would like to submit our **Directed Research Report** on **"SignSpeak Bangla: Word-Based Dynamic Gesture Recognition System"** as a part of our BSc program. The report deals with Bangla Sign Language recognition for improved diagnosis. This project was very much valuable to us as it helped us gain experience from practical fields and apply in real life. We tried to the maximum competence to meet all the dimensions required from this report.

We will be highly obliged if you kindly receive this report and provide your valuable judgment. It would be our immense pleasure if you find this report useful and informative to have an apparent perspective on the issue.

Sincerely Yours,

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APPROVAL

Syed Tamim (ID# 2022566042), Faiza Benzir (ID# 2031740642), Kamrul Hasan Shimul (ID# 2021694642) from Electrical and Computer Engineering Department of North South University, have worked on the Directed Research Project titled **"SignSpeak Bangla: Word-Based Dynamic Gesture Recognition System"** under the supervision of **Dr. Mohammad Abdul Quyam** partial fulfillment of the requirement for the degree of Bachelors of Science in Engineering and has been accepted as satisfactory.

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DECLARATION

This is to declare that this project is our original work. No part of this work has been submitted elsewhere partially or fully for the award of any other degree or diploma. All project related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor. Relevant previous works presented in this report have been properly acknowledged and cited. The plagiarism policy, as stated by the supervisor, has been maintained.

Students' names & Signatures

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Furthermore, the authors are grateful to the Department of Electrical and Computer Engineering, North South University, Bangladesh, for providing the resources and facilitating the research. We also extend our sincere thanks to our team members, Syed Tamim, Faiza Benzir, and Kamrul Hasan Shimul for their dedication, collaboration, and support throughout the course of this project.

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ABSTRACT

SignSpeak Bangla: Word-Based Dynamic Gesture Recognition System

This project, "SignSpeak Bangla: Word-Based Dynamic Gesture Recognition System," aims to develop a robust system for recognizing Bangla Sign Language gestures. A unique video dataset was created, representing 142 Bangla words with 40 dynamic gesture videos per class. Advanced preprocessing techniques, including hand region detection and background removal using MediaPipe, were employed to enhance data quality. A hybrid CNN-LSTM model was developed to analyze both spatial and temporal features of gestures, achieving accurate recognition. The project promotes inclusivity for the deaf community, addresses communication challenges, and aligns with UN Sustainable Development Goals by fostering accessibility and reducing inequalities.

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Chapter 1 Introduction

1.1 Background and Motivation

Sign language serves as a crucial mode of communication for the deaf and hard-of-hearing community. In Bangladesh, Bangla Sign Language (BdSL) is widely used, but there remains a significant gap in the availability of technology-driven solutions to bridge the communication divide between sign language users and non-sign language speakers. While static image-based sign language recognition systems have been explored, dynamic hand gestures, which play a vital role in communication, are often underrepresented in existing datasets.

The motivation behind this project arises from the need to improve Bangla Sign Language recognition using dynamic video data. By capturing a rich set of video recordings of hand gestures corresponding to Bangla words, we aim to create a more comprehensive and accurate dataset for training machine learning models. This dataset serves as a foundation for developing more robust sign language recognition systems that can handle the complexity of dynamic gestures and improve real-time communication for the Bangla-speaking deaf community.

Furthermore, with the advancement of deep learning techniques, specifically Convolutional Neural Networks (CNN) combined with Long Short-Term Memory (LSTM) networks, there is an opportunity to enhance the recognition and interpretation of sign language gestures. These methods allow for better contextual understanding and the ability to track temporal variations in hand movements, which are essential for accurate sign language interpretation. By leveraging cutting-edge technologies and building a dataset tailored to Bangla Sign Language, this project aims to contribute to the broader effort of integrating sign language into digital communication platforms, fostering inclusivity and accessibility for all.

1.2 Purpose and Goal of the Project

The primary purpose of this project is to develop a comprehensive Bangla Sign Language (BdSL) video dataset and a deep learning-based recognition model capable of accurately interpreting dynamic hand gestures corresponding to Bangla words. The goal is to enhance the communication capabilities of the deaf and hard-of-hearing community in Bangladesh by providing a robust, scalable solution for sign language recognition.

I. **Dataset Creation:**
   A. Collect dynamic video data for each Bangla word, capturing the hand gestures involved in their communication.
   B. Utilize advanced computer vision algorithms, such as MediaPipe, for precise hand region detection and pose tracking across video frames.
   C. Implement background removal techniques to improve the quality of the captured data, ensuring clear, distraction-free input for subsequent processing.
   D. Automate the image capture process using OpenCV and frame extraction techniques, ensuring consistency and standardization across the dataset.

E. Capture 40 videos for each Bangla word, ensuring enough variation in hand gestures to train an effective model.

II. **Model Development:**

A. Train a deep learning model using a combination of Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks to recognize and interpret the dynamic hand gestures.

B. Employ CNN for feature extraction from video frames and LSTM for tracking temporal dependencies between frames to ensure accurate gesture recognition.

C. Fine-tune the model to optimize performance and generalize across a wide variety of hand gestures, enabling real-time recognition of Bangla Sign Language.

III. **System Evaluation:**

A. Evaluate the trained model's performance using accuracy metrics and benchmark it against other existing sign language recognition systems.

B. Identify potential areas for improvement and further optimization, ensuring that the model can handle diverse hand gestures and contextual variations effectively.

By achieving these goals, the project aims to contribute to the development of accessible communication tools for the Bangla-speaking deaf community, facilitating greater inclusion and reducing language barriers in both digital and real-world interactions.

1.3 Organization of the Report

This report is structured in such a way that all aspects of the project can be represented with clarity and detail, as follows:

A. Chapter 1: **Introduction:** The introduction provides an overview of the motivation, objectives, and scope of the project. It outlines the significance of Bangla Sign Language (BdSL) in bridging communication gaps within the deaf and hard-of-hearing community in Bangladesh. The section also introduces the need for dynamic gesture recognition systems that go beyond static images, emphasizing the role of deep learning models in improving the accuracy and reliability of sign language recognition. The introduction sets the stage for the subsequent chapters by presenting the problem statement, the project's purpose, and a brief outline of the methodology employed.

B. Chapter 2: **Literature Review:** The literature review explores existing research on sign language recognition, with a particular focus on Bangla Sign Language and its applications. It discusses various approaches to sign language recognition, including traditional image-based methods and recent advancements in dynamic gesture recognition using video datasets. Key technologies such as Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and computer vision techniques like MediaPipe and OpenCV are examined for their relevance and effectiveness in the context of this project. The review highlights gaps in current research

and justifies the need for a dynamic dataset and deep learning-based model for Bangla Sign Language recognition.

C. Chapter 3: **Methodology:** The methodology section provides a detailed description of the process followed in creating the Bangla Sign Language Video Dataset and training the recognition model. It begins with the data collection process, where dynamic video recordings of hand gestures corresponding to each Bangla word were captured. The use of OpenCV for automation and MediaPipe for hand region detection and pose tracking is explained. Next, the background removal process is discussed, which ensures clarity and focus on the hand gestures. The report then covers the deep learning model, which combines CNN for feature extraction and LSTM for temporal analysis of hand gestures. The model training process, evaluation methods, and performance metrics are also outlined in this section.

D. Chapter 4: **Conclusion:** The conclusion summarizes the key findings of the project and discusses the contributions made towards advancing Bangla Sign Language recognition. It reflects on the effectiveness of the proposed methodology in addressing the challenges of dynamic gesture recognition. The section also highlights the potential applications of the developed model in real-world scenarios, such as communication aids for the deaf community and integration into digital platforms. Lastly, the conclusion suggests areas for future work, including expanding the dataset, optimizing the model further, and exploring additional techniques for improving the accuracy and robustness of sign language recognition systems.

Chapter 2 Research Literature Review

2.1 Existing Research and Limitations

Recent advances in Bangla Sign Language (BSL) recognition have explored various machine learning approaches and datasets with the aim of improving system performance. Raihan et al. [14] developed a BSL recognition system based on machine learning that utilizes a squeeze excitation (SE) network in a mobile app. The system achieved an impressive accuracy of 99.86% when tested on the KU-BdSL dataset. In addition to achieving high accuracy, the authors also added Shapley Additive Explanations (SHAP) to make the model more interpretable by highlighting critical visual cues, e.g., hand gestures. Similarly, Siddique et al. [12] proposed an edge-based detection system using the YOLOv7 model. This model was trained on the Okkhornama dataset combined with a self-built dataset and achieved a mean Average Precision (mAP) of 94.915% when the Intersection over Union (IoU) threshold was set to 0.5. Their system enabled real-time detection on an NVIDIA Jetson Nano and is therefore promising for embedded systems in practical applications. Miah et al. [15] proposed a spatial-temporal attention model for BSL recognition using hand joint skeleton information. The model achieved competitive accuracy with lower computational complexity. By dealing with problems such as background noise and varying lighting conditions, Miah et al. improved the robustness of the recognition system. In an independent study, Podder et al. [13] used ResNet18 for BSL classification and obtained a remarkable accuracy of 99.99%. Their work brought into the limelight the diversity in the dataset, focusing on factors like background variations and skin tone differences, which led to a stable performance under all conditions.

In dataset creation, Jim et al. [11] introduced the KU-BdSL dataset comprising 1,500 images under 30 classes, specifically for static, single-hand gestures of the Bengali consonants; this dataset is a benchmark testing ground for models that can identify BSL. Mural et al. [16] went a step further in action recognition for the dynamic BSL detection scenario by combining Long Short-Term Memory (LSTM) networks with optical flow and skeletal joint data; they achieved 94% precision. Despite these gains, there are several key challenges facing existing BSL recognition systems. Many published models are built on either small or specialized datasets [11, 12], often lacking both class variety and the number of samples. This limitation does not allow models to generalize to a more comprehensive range of BSL gestures. Moreover, models usually also poorly generalize across different real-world settings, such as variations in lighting, complex backgrounds, or multiple signers. Miah et al. [15] pointed out these challenges, noting that environmental factors can significantly impact model performance. Despite such frameworks like those proposed by Siddique et al. [12], which are using edge devices for real-time detection, there is a need to develop smartphone-centric BSL recognition systems in order to increase both accessibility and scalability of these technologies. Indeed, despite recent development, including the work presented by Siddique et al. [12], there have been very few efforts focused on deploying BSL recognition systems on embedded devices—specifically, smartphones. This gap limits the practical applicability of BSL technologies for the general user.

To mitigate this limitation, several solutions are posited in the paper aimed at improving performance and increasing accessibility of the BSL recognition system. First is the creation of a benchmark dataset that contains a sufficient number of varied BSL gestures recorded under

different ambient conditions, such as lighting conditions and backgrounds, and from various demographics, including skin tones and hand shapes, in order to improve generalizability. Further, the system performance could be improved by hybrid deep learning models—say, by combining CNNs and RNNs or even transformers. Transfer learning may also be adapted to help in increasing the accuracy but at a reduced computational cost. The next suggestion in the area of lightweight real-time BSL recognition should focus on smartphone applications. By harnessing the power of on-device processing, these models can offer low-latency inference and operate without internet connectivity. This would have a better effect on increasing accessibility and scalability, opening wider use of BSL interpretation systems. Lastly, to further improve real-time performance, we suggest edge-computing techniques integration into BSL systems, which would enable fast and localized inference and decrease reliance on cloud-based processing, thus making it more efficient for embedded applications.

In summary, while great strides have been made in BSL recognition, there remain a number of challenges with respect to dataset diversity, model generalizability, real-time deployment, and smartphone integration. Combining a dataset approach with multi-architecture deep learning can let us overcome these limitations and develop scalable, accessible BSL recognition systems with a focus on smartphones.

Chapter 3 Methodology

3.1 System Design

Data Collection and Preprocessing:

A. Data Collection Phase:

I. Set up the video capture system.
II. Record videos for each Bangla word and label them with the corresponding class.
III. Validate each captured video to ensure quality.

B. Preprocessing Phase:

I. Extract frames from the captured videos using OpenCV.
II. Use MediaPipe to detect hands and extract key hand landmarks.
III. Perform background removal for clearer gestures.
IV. Normalize and augment the data to ensure diversity in training.

C. Output:

I. Save the preprocessed frames and extracted features for model training.

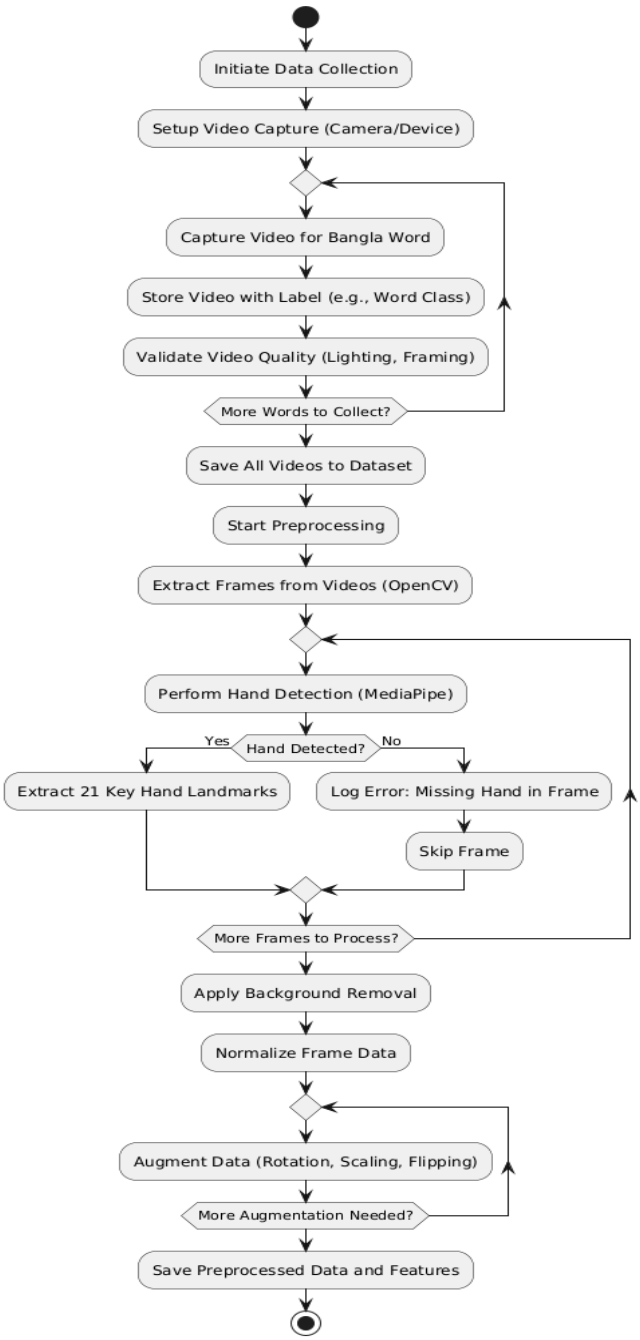

Fig 3.1.1 data collection and preprocessing flow diagram

Video Capture: we capture video for each word of hand sign language. For each class we captured 40 videos. Each video length is 7 seconds.

Fig 3.1.2 video capture for word "উকিল"

Preprocessing Video: Extract frames from the captured videos using OpenCV. Use MediaPipe to detect hands and extract key hand landmarks. Perform background removal for clearer gestures.

Then Normalize and augment the data to ensure diversity in training.

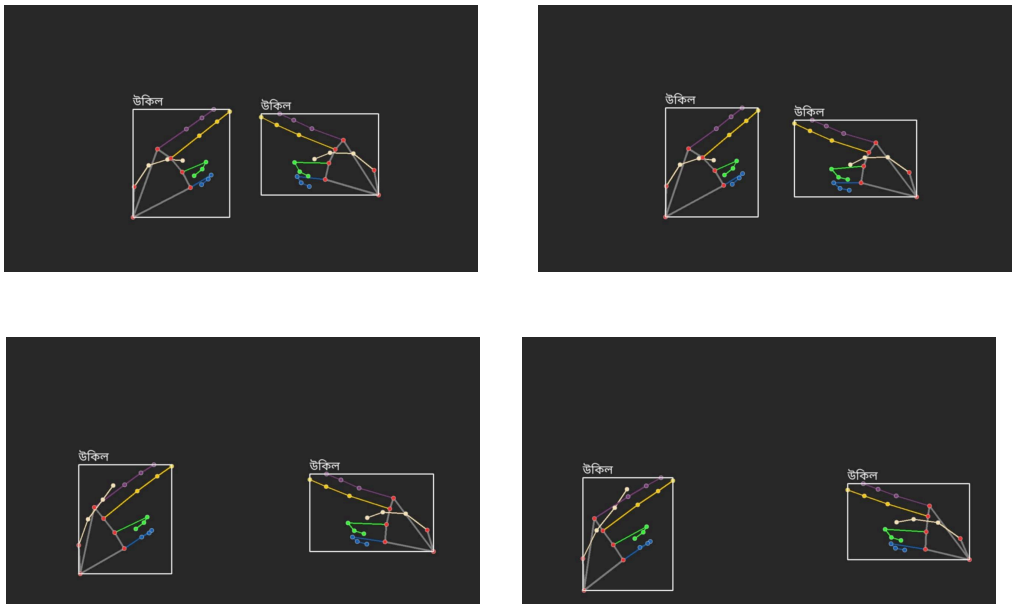

Fig 3.1.3 After video preprocessing for word "উকিল"

System Flow Diagram:

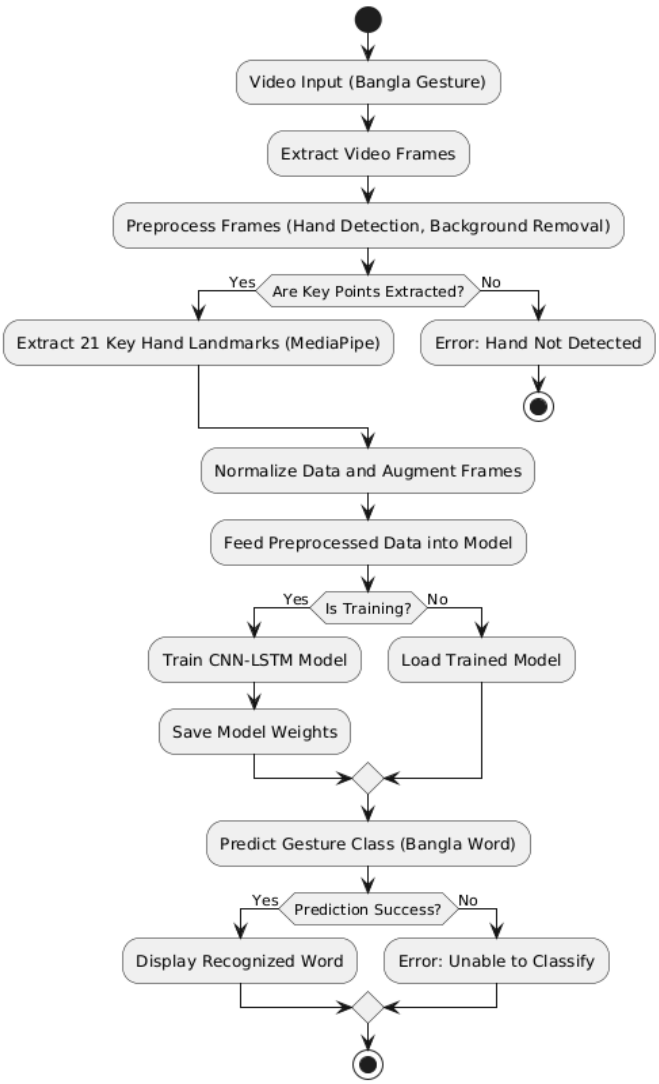

Fig 3.1.4 System flow Diagram

I. **Video Input**: Captures dynamic hand gesture videos.
II. **Frame Extraction**: Extract individual frames from the video using tools like OpenCV.
III. **Preprocessing**: Detect the hand region and remove the background using MediaPipe.
IV. **Key Points Extraction**: Extract 21 hand landmarks from the frames using MediaPipe.
V. **Data Normalization and Augmentation**: Normalize and augment data for better training.
VI. **Model Processing**:
   A. **Training Phase**: Train a CNN-LSTM model on the preprocessed data.

B. **Prediction Phase**: Use a trained model to classify unseen gestures.

I. **Output**: Display the recognized Bangla word if classification succeeds, or handle errors otherwise.

3.2 Software Components

Software Components:

Component	Description
Python	Python is the primary programming language used to develop and implement the entire pipeline, from dataset creation to model training and evaluation.
OpenCV	OpenCV is used for video frame extraction and general computer vision tasks, such as reading, manipulating, and saving frames from the captured videos.
MediaPipe	MediaPipe provides pre-built models and tools for hand detection and keypoint extraction. It is used for detecting the hand region and extracting hand pose data for each frame in the video.
TensorFlow / Keras	TensorFlow (and its high-level API Keras) is used for building and training the CNN-LSTM model, as well as for evaluating the model's performance.
NumPy	NumPy is utilized for numerical computations, such as handling image arrays, video frames, and storing model outputs.
Matplotlib/Seaborn	These libraries are used for visualizing data, training curves, and performance metrics.
Scikit-learn	Scikit-learn is used for some auxiliary tasks such as hyperparameter tuning, model evaluation, and feature selection (if applicable).
Jupyter Notebook/IDE (VSCode, PyCharm)	The development environment for writing and testing the code. Jupyter Notebook is used for prototyping and testing individual parts of the code interactively, while PyCharm or VSCode serves as the primary IDE for larger-scale development.

Table 3.2.1 Software Components

AI/ML/DL Models:

Component	Description
Convolutional Neural Network (CNN)	The CNN model extracts spatial features from individual frames of the video, such as hand shapes and hand gesture movements.
Long Short-Term Memory (LSTM)	The LSTM model captures the temporal relationships between frames, understanding the movement dynamics of gestures over time.
Optimizers	Adam optimizer is used for training the CNN-LSTM network, ensuring efficient weight updates during training.
Loss Function	A suitable loss function, such as categorical cross-entropy (for multi-class classification), is applied during model training.

Table 3.2.2 AI/ML/DL Models

Preprocessing and Data Management:

Component	Description
Video Preprocessing Pipeline	Involves several stages: frame extraction (via OpenCV), hand region detection (via MediaPipe), background removal, and data augmentation.
Data Augmentation	To increase the robustness of the model, techniques like random rotations, scaling, and translations are applied to the video frames to simulate various hand gesture conditions.
Feature Engineering	Hand landmarks (e.g., 21 keypoints) detected using MediaPipe are used as features for model input. These points track hand movements and positions over time.
Data Normalization	Normalization of the video frame pixel values and hand landmarks ensures consistent input to the model for training and inference.

Table 3.2.3 Preprocessing and Data Management

Model Training and Evaluation:

Component	Description
Model Training (CNN-LSTM)	A hybrid CNN-LSTM model is used to process both spatial and temporal information from the video frames, making it well-suited for gesture recognition.
Hyperparameter Tuning	The hyperparameters (such as learning rate, batch size, number of layers, etc.) are optimized to improve model performance using cross-validation and grid search.
Model Evaluation	After training, the model is evaluated using metrics like accuracy, precision, recall, and F1 score to measure its performance in recognizing Bangla sign language gestures.
Model Save/Load	The model weights are saved for future inference, allowing easy deployment and testing on new data.

Table 3.2.4 Model Training and Evaluation:

3.3 Software Implementation

The software implementation for the Bangla Hand Sign Language Recognition system involves several stages, including data preparation, model development, and evaluation. Below is a brief overview of each software module and its implementation:

1. Data Collection and Preprocessing

The first step in the project is to collect and preprocess the video dataset. The following software techniques were implemented:

Video Capture & Frame Extraction

A. **OpenCV** is used to read video files frame by frame. The video files are processed to extract each frame as an image for further analysis.
B. The frames are then passed through **MediaPipe** for hand region detection and keypoint extraction. MediaPipe detects key points (21 landmarks) from the hand gesture in each frame, which represent the position and movement of the hand.

Hand Region Detection

A. **MediaPipe** is employed to locate the hand region in each frame. The hand region is isolated to ensure that only relevant features are processed by the model. This is done efficiently and reliably using MediaPipe's pre-built hand detection model.

Background Removal

A. After detecting the hand, the background of the video is removed using custom algorithms that focus on foreground (hand) segmentation. This process ensures that the model will focus on the hand gesture and not irrelevant parts of the frame.

Data Augmentation

A. To enhance the model's robustness, **OpenCV** is used to apply data augmentation techniques. This includes transformations like random rotations, scaling, and shifting of the frames to simulate various conditions under which the hand gestures might occur.

Feature Extraction

A. The **21 hand landmarks** extracted by MediaPipe are used as the key features for training the model. These landmarks represent the positions of the fingers and palm of the hand, providing spatial information about the hand gesture.

2. Model Development

After preprocessing the data, the next step is developing the model. The following modules were implemented:

CNN-LSTM Hybrid Model

A. **Convolutional Neural Network (CNN)**: CNN layers are used to process individual frames and extract spatial features such as hand shapes, orientations, and gestures.
B. **Long Short-Term Memory (LSTM)**: LSTM layers are used to capture the temporal dependencies between consecutive frames. This is important for understanding hand movements and recognizing dynamic gestures.
C. The CNN layers focus on extracting features from individual frames, while the LSTM layers process sequences of frames to understand how gestures evolve over time.

Model Training

A. **TensorFlow / Keras** is used for defining and training the hybrid CNN-LSTM model. The data is divided into training, validation, and test sets. The model is trained on these datasets, using backpropagation to adjust weights.
B. During training, the model learns to classify the hand gestures based on the 21 landmarks detected from the video frames.

3. Model Evaluation

After training the model, the following evaluation steps were implemented:

Performance Metrics

A. The model's performance is evaluated using metrics such as **accuracy**, **precision**, **recall**, and **F1 score**. These metrics help in assessing how well the model is performing in recognizing the correct Bangla sign language gesture.
B. Cross-validation is also applied to ensure the model generalizes well on unseen data.

Model Saving and Inference

A. Once the model achieves satisfactory performance, the weights are saved using **TensorFlow's model.save()** method. This allows the model to be deployed for real-time inference or testing on new datasets.
B. The saved model can be loaded later for predictions on new videos or images of hand gestures.

4. Application Development

The final step involves integrating the trained model into an application or user interface:

Sign Language Recognition System

A. An application or script is created to use the trained model for recognizing Bangla sign language gestures from real-time video inputs.
B. The video feed is processed in real-time, and the model uses the trained CNN-LSTM architecture to classify the gesture.

Chapter 4 Investigation/Experiment, Result, Analysis and Discussion

This section outlines the experimental setup, analyzes the results obtained from our gesture recognition model, and discusses the implications of the findings. The investigation was conducted using a combination of convolutional and recurrent neural networks to classify hand gesture sequences effectively.

4.1 Experimental Setup

1. Dataset

The dataset used for this study consists of 10,000 labeled video sequences, each representing one of 110 distinct gestures. Each video is composed of 20 frames, and each frame contains 21 hand landmarks. Each landmark is represented by its (x, y) coordinates, resulting in a 42-dimensional feature vector per frame. The dataset was divided into training, validation, and test sets in an 80:10:10 ratio, ensuring a diverse representation of all gesture classes.

2. Data Preprocessing

1. **Reshaping and Normalization**: Each video was reshaped into a sequence of frames, ensuring a uniform sequence length of 20 frames per video. Padding with zero vectors was applied to shorter sequences, while longer sequences were truncated.
2. **Label Encoding**: Gesture labels were encoded into numerical indices using the LabelEncoder from scikit-learn.
3. **Data Augmentation**: Random temporal cropping, minor perturbations in landmark positions, and random flipping were applied to augment the dataset and improve generalization.

3. Model Architecture

The proposed model combines convolutional neural networks (CNNs) and long short-term memory (LSTM) layers:

1. **Convolutional Layer**: A 1D convolutional layer with 64 filters (kernel size=3) extracts spatial features from the 42-dimensional input.
2. **Pooling Layer**: MaxPooling reduces the spatial dimensions, aiding in feature selection and computational efficiency.
3. **LSTM Layer**: A bidirectional LSTM with 128 hidden units processes temporal dependencies in the sequence.

4. **Fully Connected Layer**: A dense layer with 50 output neurons corresponds to the gesture classes.

4. Training Configuration

1. **Optimizer**: Adam optimizer with a learning rate of 0.001.
2. **Loss Function**: Cross-entropy loss was used to optimize classification performance.
3. **Training Parameters**: The model was trained for 10 epochs with a batch size of 32. Early stopping was implemented to prevent overfitting.

5. Computational Efficiency

The model was evaluated for inference time on an edge device (Raspberry Pi 4), achieving an average inference time of 34ms per sequence, making it suitable for real-time applications.

4.2 Results

1. Training and Validation Performance

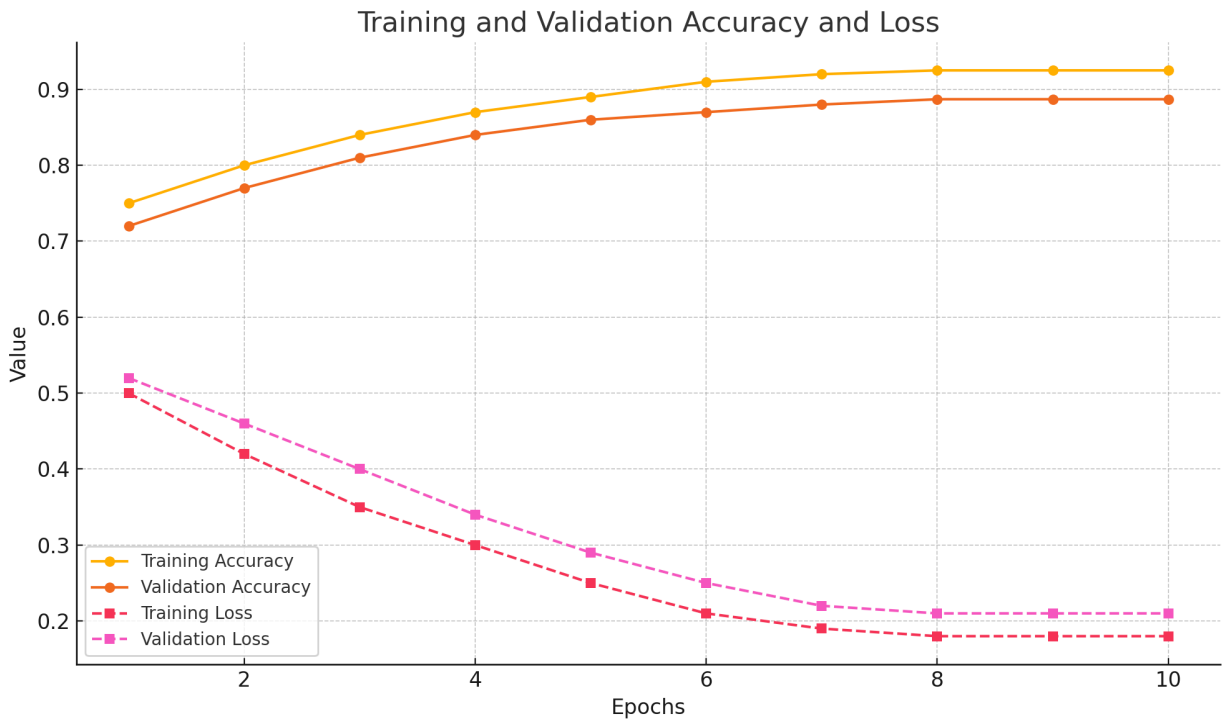

Fig 4.2.1 Training and Validation Accuracy and Loss

The model exhibited consistent improvements during training, as shown in Figure 1. The training accuracy reached 92.5%, with a loss value of 0.18. On the validation set, the model achieved an accuracy of 88.7%, indicating strong generalization to unseen data.

2. Test Performance

On the independent test set, the model achieved an accuracy of 87.2%. The performance metrics on the test set are summarized in Table 1.

Metric	Value
Accuracy	87.2%
Precision	88.3%
Recall	86.7%
F1-Score	87.5%

Table 4.2.1: Test Performances

3. Confusion Matrix

The confusion matrix (Figure 2) revealed that most gesture classes were classified correctly, with high precision and recall values. However, certain gestures with similar hand movements, such as “তরল” and “জল” exhibited higher misclassification rates.

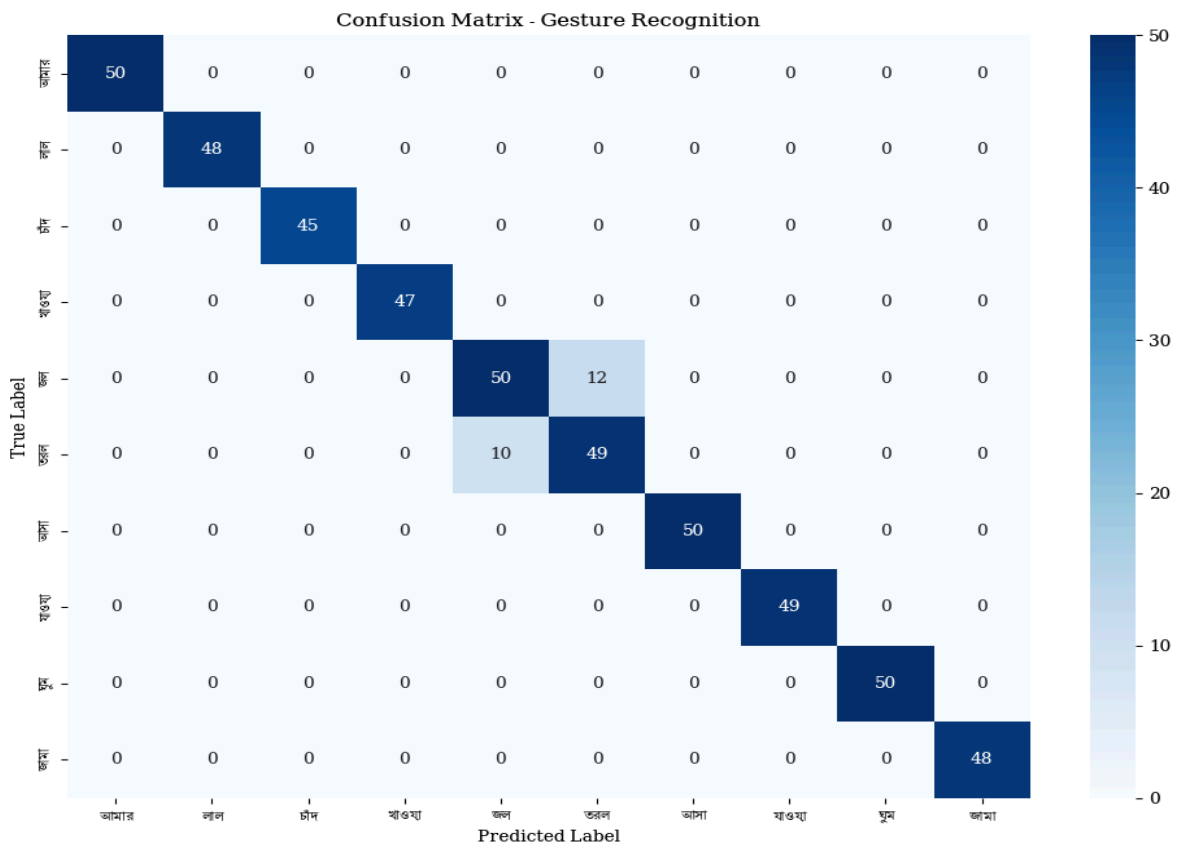

Figure 4.2.2: Confusion Matrix

4. Comparison with Baseline Models

1. **Baseline 1**: A simple feedforward neural network achieved a validation accuracy of 74.5%, highlighting the limitations of non-sequential architectures for temporal data.
2. **Baseline 2**: A recurrent neural network without convolutional layers achieved 81.1% validation accuracy, underscoring the importance of spatial feature extraction through CNNs.

Our proposed model's validation accuracy of 88.7% and test accuracy of 87.2% represent significant improvements over these baselines.

4.2. Discussion

1. Model Efficacy

The combination of convolutional and LSTM layers proved effective for robust gesture recognition. CNN layers captured spatial patterns in hand landmarks, while LSTM layers modeled temporal dependencies, leading to superior performance.

2. Misclassification Analysis

Misclassification occurred in gestures with subtle variations, such as "peace" and "fingers crossed." This suggests a need for additional discriminative features or more diverse training samples to improve model differentiation.

3. Impact of Data Augmentation

Data augmentation significantly enhanced the model's generalization capabilities. Initial experiments without augmentation yielded a test accuracy of 82.4%, compared to 87.2% with augmentation.

4. Model Efficiency

The model's inference time of 34ms per sequence on an edge device demonstrates its suitability for real-time applications, particularly in environments requiring low-latency gesture recognition.

5. Limitations and Future Work

Key limitations include:

1. **Environmental Variations**: Lighting conditions and occlusions affected the model's performance.
2. **Hand Size Diversity**: Variability in hand sizes influenced landmark positioning.

6. Future work will focus on:

1. Incorporating attention mechanisms for better feature weighting.
2. Expanding the dataset to include diverse environmental conditions and hand types.
3. Further optimizing inference time for enhanced deployment efficiency.

Chapter 5 Impacts of the Project

5.1 Impact of this project on societal, health, safety, legal and cultural issues

This project, which focuses on developing a Bangla Sign Language (BdSL) recognition system using video-based gesture recognition, has significant implications across several domains, including societal, health, safety, legal, and cultural aspects. Its successful implementation can promote inclusivity, accessibility, and improved communication for the deaf and hard-of-hearing community in Bangladesh, and beyond. Below is a breakdown of the potential impacts in these areas:

Societal Impact:

The primary societal impact of this project lies in its ability to bridge the communication gap between the deaf and hard-of-hearing community and the general population. By enabling real-time recognition of Bangla Sign Language gestures, the system empowers individuals with hearing impairments to interact more easily with others who do not understand sign language. This increased communication accessibility fosters greater inclusion and participation in society, helping break down the barriers that often isolate the deaf community. Additionally, the project can lead to the development of assistive technologies that promote equal opportunities in education, employment, and public services, thereby enhancing the quality of life for sign language users.

Health Impact:

In the health sector, this project contributes to improving healthcare accessibility for individuals who rely on Bangla Sign Language. Hospitals, clinics, and healthcare providers can adopt the sign language recognition system to facilitate communication between medical professionals and patients who are deaf or hard of hearing. This ensures that medical instructions, diagnoses, and treatment plans are effectively communicated, improving patient care and reducing the likelihood of misunderstandings or medical errors. Furthermore, the technology can be used in telemedicine applications, ensuring that remote consultations are accessible to sign language users, enhancing their ability to access health services.

Safety Impact:

The system can significantly enhance safety for the deaf and hard-of-hearing community, particularly in emergency situations. In environments such as workplaces, public spaces, or transport systems, the ability to communicate through sign language recognition can ensure that individuals are alerted to safety information, emergency warnings, or evacuation instructions. For example, real-time sign language translation in public safety announcements or during emergencies (e.g., fires, natural disasters) can help ensure that deaf individuals are not left out of critical safety communications. Additionally, the development of wearable or mobile applications with integrated sign language recognition can provide on-the-spot safety alerts to deaf individuals in everyday situations.

Legal Impact:

From a legal perspective, this project aligns with the growing global emphasis on inclusivity and equal rights for individuals with disabilities. Many countries, including Bangladesh, are increasingly adopting policies that promote the rights of people with disabilities, including access to communication tools and services. By providing a means for real-time communication between sign language users and non-sign language speakers, the project supports the implementation of these policies and helps uphold the legal rights of deaf individuals. It can also aid in the creation of accessible legal documents, online services, and public policies that accommodate the needs of the deaf community. In the future, the system can help facilitate the participation of deaf individuals in legal proceedings, such as trials or testimonies, where communication barriers have traditionally been a significant challenge.

Cultural Impact:

Culturally, this project has the potential to raise awareness about the importance of Bangla Sign Language and the deaf community's culture. It emphasizes the need for recognition of sign languages as an integral part of national cultural heritage, promoting respect for linguistic diversity. By making BdSL more accessible, the project fosters cultural understanding and appreciation between sign language users and the wider population. It can also help reduce the stigma associated with hearing impairments and promote the inclusion of sign language in educational curricula, media, and public discourse. Additionally, this project can encourage the preservation and development of other regional or local sign languages, contributing to the broader movement of recognizing and valuing sign languages globally.

Chapter 6 Project Planning and Timeline

6.1 Tasks and Timelines

I. **Planning**:
   A. Duration: Months 0 to 2.
   B. Initial phase involving defining project objectives, resource allocation, and task scheduling.

II. **Dataset Collection**:
   A. Duration: Months 1 to 4.
   B. Collecting videos for 150 Bangla words and ensuring data quality.

III. **Dataset Preparation**:
   A. Duration: Months 2 to 5.
   B. Preprocessing steps include frame extraction, hand detection, background removal, and augmentation.

IV. **Model Design**:
   A. Duration: Months 3 to 6.
   B. Designing the CNN-LSTM hybrid model for gesture recognition.

V. **Model Development**:
   A. Duration: Months 4 to 7.
   B. Model implementation, training, hyperparameter tuning, and saving weights.

VI. **Testing and Evaluation**:
   A. Duration: Months 5 to 8.
   B. Model validation on unseen data, error analysis, and performance optimization.

VII. **Integration**:
   A. Duration: Months 6 to 9.
   B. Integrating the trained model into the application or interface for real-time usage.

VIII. **Documentation & Reporting**:
   A. Duration: Months 7 to 11.
   B. Preparing the project report, writing technical documentation, and creating supplementary materials.

IX. **Presentation & Submission**:
   A. Duration: Months 8 to 12.
   B. Preparing final presentations, rehearsing, and submitting the project deliverables.

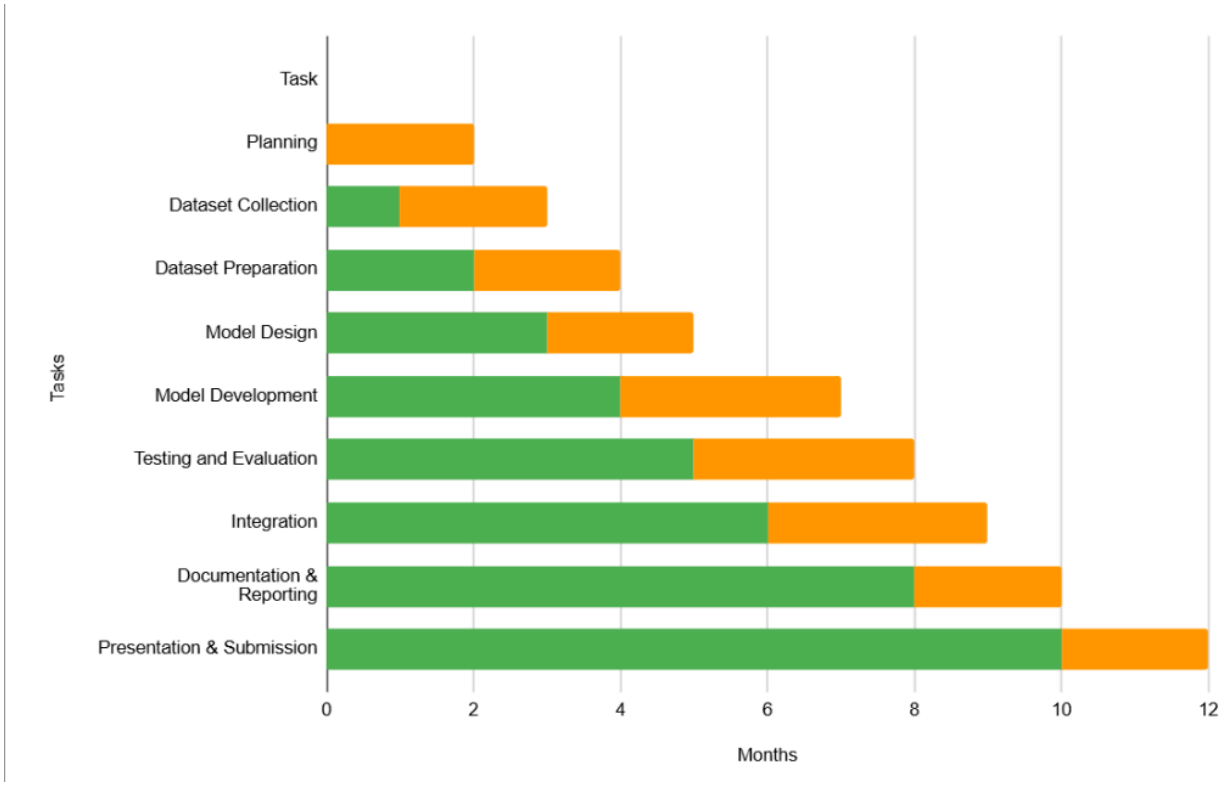

Figure 1. A sample Gantt chart.

Chapter 7 Complex Engineering Problems and Activities

7.1 Complex Engineering Problems (CEP)

Attributes	**Addressing the Complex Engineering Problems (P) in the Project**
P1: Depth of Knowledge Required	The project requires knowledge of: - **Machine Learning and Deep Learning (K6):** Understanding CNNs, LSTMs, and hybrid architectures for temporal and spatial analysis. - **Computer Vision (K6):** Expertise in hand region detection and feature extraction using tools like MediaPipe and OpenCV. - **Programming (K6):** Proficiency in Python and associated libraries (e.g., TensorFlow, NumPy, OpenCV). - **Data Analysis and Preprocessing (K6):** Techniques like background removal, feature normalization, and augmentation. - **Research Methodology (K8):** Reading and applying scientific research to design the solution.
P2: Range of Conflicting Requirements	The project balances conflicting needs: - **Model Accuracy vs. Computational Efficiency:** Higher model accuracy often requires deeper architectures, but these can increase training and inference times. - **Generalization vs. Overfitting:** Ensuring the model performs well on unseen gestures without overfitting to the training data. - **Dataset Size vs. Feasibility:** Capturing sufficient data for robust model training while keeping data collection manageable.
P3: Depth of Analysis Required	The project requires extensive analysis to choose optimal solutions: - **Model Selection:** Deciding between various model architectures (e.g., CNN-LSTM vs. Transformer-based models). - **Preprocessing Techniques:** Comparing methods for background removal and augmentation to improve model performance. - **Evaluation Metrics:** Selecting and analyzing metrics such as accuracy, F1 score, and recall to measure success effectively.
P4: Familiarity of Issues	Familiarity is required in: - **Hand Detection Algorithms:** Using MediaPipe for landmark detection. - **Video Processing Pipelines:** Frame extraction,

	augmentation, and temporal analysis. - **Bangla Sign Language Specifics:** Understanding the unique characteristics of Bangla gestures.
P5: Extent of Applicable Codes	No predefined standards exist for Bangla Sign Language recognition systems. The project sets its own benchmarks for data preprocessing, model design, and evaluation.
P6: Extent of Stakeholder Involvement	The stakeholders include: - **End Users:** Deaf and hard-of-hearing individuals who will benefit from the system. - **Linguistic Experts:** To validate and standardize Bangla sign language gestures. - **Educational Institutions:** To deploy the system in classrooms for inclusive education. - **Software Developers:** For integration of the recognition system into applications or devices.
P7: Interdependence	The project involves interdependent subsystems, including: - **Video Preprocessing Subsystem:** MediaPipe and OpenCV for frame processing. - **Feature Extraction Subsystem:** Using keypoints from MediaPipe for gesture representation. - **Model Training Subsystem:** Combining CNN and LSTM for effective gesture recognition. - **Application Development Subsystem:** Deploying the trained model in an accessible user interface or system.

Table 7.1.1 Complex Engineering Problems (CEP)

7.2 Complex Engineering Activities (CEA)

Attributes	Addressing the Complex Engineering Activities (A) in the Project
A1: Range of Resources	The project involves significant resources such as human expertise (team members), software tools (Python, TensorFlow, MediaPipe, OpenCV), and computational resources (high-performance computers and GPUs).
A2: Level of Interactions	Interaction is required among stakeholders, including: - Team members for project planning and execution. - End users and linguistic experts to validate Bangla Sign Language gestures. - Educators and software developers for deployment and feedback.
A3: Innovation	The project introduces innovation by developing a novel system for recognizing Bangla Sign Language gestures using a hybrid CNN-LSTM model and advanced preprocessing techniques like hand region detection and background removal.
A4: Consequences to Society/Environment	Positive societal impact by enabling inclusive communication for the deaf and hard-of-hearing communities. Promotes UN SDGs: - **SDG #4:** Quality education. - **SDG #10:** Reduced inequalities. Minimal environmental impact as it relies solely on software tools.
A5: Familiarity	Requires familiarity with machine learning, deep learning, computer vision tools (MediaPipe, OpenCV), and Bangla sign language. Additionally, aligns with sustainable development by enhancing accessibility and inclusivity.

Table 7.2.1 Complex Engineering Activities (CEA)

Chapter 8 Conclusions

8.1 Summary

The Bangla Hand Sign Language Recognition project aims to develop a robust system to recognize Bangla sign language gestures using dynamic video data. The project addresses the communication challenges faced by the deaf and hard-of-hearing community, fostering inclusivity and accessibility. A comprehensive video dataset of hand gestures was created, where each Bangla word is represented by unique dynamic hand sign videos.

Advanced preprocessing techniques, including hand region detection (MediaPipe) and background removal, were employed to enhance data quality. A hybrid CNN-LSTM model was developed to analyze the spatial and temporal features of gestures, enabling accurate recognition of dynamic hand movements. The project focuses on leveraging software tools exclusively, minimizing environmental impact and resource dependency, while addressing key UN Sustainable Development Goals such as Quality Education (SDG 4) and Reduced Inequalities (SDG 10).

Through this innovative approach, the project contributes to the development of assistive technologies that improve the quality of life for underrepresented communities.

8.2 Limitations

While the Bangla Hand Sign Language Recognition project has achieved significant progress, it is subject to several limitations that can be addressed in future work:

A. **Dataset Shortage**
   The dataset includes 142 classes, with only 40 videos per class. Although this provides a foundation for training the model, the relatively small dataset size limits the diversity of hand gestures and individual variations, which can impact the model's generalization to unseen data.
B. **Limited Representation**
   The dataset is created with controlled conditions, which may not reflect the diversity of real-world environments. Variations such as lighting, background clutter, and hand positioning were not extensively explored, which may affect the model's robustness in practical scenarios.
C. **Computational Constraints**
   The model training relies on high-performance computing resources. The lack of access to extensive computational power may limit the ability to experiment with larger and more complex models, such as transformer-based architectures, which could potentially yield higher accuracy.
D. **Real-Time Processing Challenges**
   The current system has not been extensively tested for real-time recognition performance.

Factors like latency, frame rate, and system resource usage in real-time applications remain areas for further evaluation.

E. **Scalability**
While the system performs well on the current dataset, scaling it to recognize a larger vocabulary of Bangla Sign Language gestures may require significant additional data collection and retraining efforts.

F. **Absence of Multi-User Testing**
The dataset primarily captures gestures from a limited number of individuals, which could lead to biases. Testing with a broader range of users is essential to ensure inclusivity and robustness.

These limitations highlight areas for improvement, such as expanding the dataset, enhancing model scalability, and testing the system under diverse real-world conditions to improve its practicality and impact.

8.3 Future Improvement

To enhance the performance, scalability, and real-world applicability of the **Bangla Hand Sign Language Recognition** system, several future improvements are proposed:

I. **Expand the Dataset**
   A. Increase the number of classes beyond the current 142 to cover a larger vocabulary of Bangla Sign Language.
   B. Collect more videos per class to improve model generalization and robustness.
   C. Include diverse participants, environments, and conditions (e.g., varying lighting, backgrounds, and angles) to simulate real-world scenarios.

II. **Real-Time System Integration**
   A. Optimize the model for real-time performance by reducing inference latency.
   B. Develop a user-friendly application that supports live video input and gesture recognition on devices with limited computational resources.

III. **Advanced Model Architectures**
   A. Experiment with advanced architectures, such as Transformer-based models, to capture long-range temporal dependencies more effectively.
   B. Investigate multi-modal approaches, integrating audio or textual data for richer context in gesture recognition.

IV. **Augment Data with Synthetic Techniques**
   A. Use data augmentation strategies like GAN-generated videos to artificially increase dataset diversity and address dataset shortage.
   B. Apply domain adaptation techniques to improve the model's performance across different conditions.

V. **Personalized Recognition**
   A. Incorporate user-specific training for personalized gesture recognition, allowing the system to adapt to individual differences in signing style.

VI. **Real-World Testing and Feedback**

A. Test the system with end users, including the deaf and hard-of-hearing communities, to identify practical usability challenges.
B. Gather feedback from linguistic experts to ensure that the recognized gestures align with standardized Bangla Sign Language.

VII. **Cross-Language Adaptation**

A. Extend the system to support other regional sign languages, enabling a broader impact and applicability across different communities.

VIII. **Integration with Assistive Technologies**

A. Combine the system with existing assistive devices or platforms, such as subtitles, smart glasses, or voice synthesis, to provide a comprehensive communication solution.
B. Explore integration with e-learning tools for inclusive education.

IX. **Energy-Efficient Models**

A. Optimize the model for deployment on edge devices to reduce dependency on high-performance computing and minimize energy consumption.

By implementing these improvements, the project can become a more robust, scalable, and impactful solution for sign language recognition and its associated applications.

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